

Monte Carlo Options Pricing System

M7 Checkpoint — React Frontend Dashboard

Design Notes

August 2026

1 Overview

M7 delivers an interactive React dashboard that drives the full Monte Carlo workflow through the M6 FastAPI backend. The frontend is a demo vehicle for the computational core, not a production UI; decisions favoured simplicity and fast iteration over polish.

2 Stack

- **Build:** Vite + React 19 (JavaScript, not TypeScript)
- **Styling:** Tailwind CSS v4 via `@tailwindcss/vite` (no config file)
- **Charts:** Recharts 3
- **HTTP:** axios
- **Routing:** react-router-dom 7
- **Icons:** lucide-react
- **State:** React hooks + context only (no Redux/Zustand)

3 Architecture Decisions

1. **Vite + React over Next.js** — no SSR, no server routing needed; the dashboard is a pure SPA consuming a REST API.
2. **Four pages, not five** — Simulate and Price combined into one page because they form a natural sequential workflow; splitting them adds clicks with no benefit.
3. **No state management library** — each page manages its own API calls via custom hooks; a small `AppContext` shares ticker, calibrated params, and S_0 across pages.
4. **Dark theme** — quantitative finance aesthetic; dark gray backgrounds (`#0f1117`), lighter card surfaces (`#1a1d27`), blue/teal accent.

4 Pages

4.1 Calibrate

Pick ticker → set lookback days (default 756) → fire 3 parallel `POST /calibrate` calls (GBM, JD, Heston) → display results in a 3-column grid. Each card shows calibrated params, fit metrics (log-likelihood, AIC, BIC), and a Feller condition badge for Heston. On success, params stored in `AppContext`.

4.2 Simulate & Price

Simulate section: model selector, $S_0/T/dt/ n_{\text{sim}}/\text{variance reduction}/\text{seed}$ inputs. Confidence cone chart (Recharts range areas for percentile bands 5/25/50/75/95 + sample paths). Summary stats below.

Price section: option type dropdown (8 payoffs), strike K , r , q , barrier (conditional). Reuses simulation config. Shows price $\pm$ SE, 95% CI, BS benchmark + relative error, payoff histogram (bar chart from pre-bucketed API data), Greeks table (Delta, Gamma, Vega, Theta, Rho).

4.3 Risk

Toggle between single-asset and portfolio mode. Single-asset: model + params + optional scenario from `GET /risk/scenarios` $\rightarrow$ `POST /risk/metrics`. Portfolio: add multiple (ticker, weight, model_type) rows $\rightarrow$ `POST /risk/portfolio`. Both display VaR, CVaR, max draw-down, Sharpe, $P(\text{loss})$. Scenario comparison shows baseline vs stressed side by side.

4.4 Runs

Lightweight MLflow browser: experiment name, ID, run count. Links to the full MLflow UI at `localhost:5000`.

5 Shared Context

`AppContext` provides:

- `selectedTicker` — set on Calibrate, consumed by other pages
- `calibratedParams` — `{gbm, jump_diffusion, heston}` objects; cleared on ticker change to prevent cross-ticker param leakage
- `S0` — latest price, pre-fills simulation inputs

6 Implementation Notes

- **Confidence cone:** built as Recharts range areas (`[low, high]` pairs), not stacked areas — stacked areas leak a zero baseline into the y-domain and flatten the plot.
- **Parameter display:** `dt` and model params rounded for display (params to 6dp in editable forms); Calibrate page shows raw API values.
- **Extra files added by Claude Code:** `lib/models.js` (model field definitions + payoff argument rules), `lib/format.js`, `components/ui.jsx` (Card/Field/Button primitives), `components/ParamInputs.jsx`, `hooks/useApiAction.js`, `hooks/useModelParams.js`.
- **Ticker change clears params** — silently carrying one ticker's calibrated volatility into another ticker's pricing is a dangerous silent error.

7 Backend Change

`POST /risk/metrics` was missing scenario support (only `/risk/portfolio` had it). Added `scenario` and `custom_scenario` fields to `RiskMetricsRequest`, applied server-side via `apply_scenario()`. 5 new tests. 487 backend tests now pass (was 482).

8 Files

`frontend/src/`

```
App.jsx
main.jsx
index.css
api/
  client.js
context/
  AppContext.jsx
hooks/
  useCalibration.js
  useSimulation.js
  usePricing.js
  useRisk.js
  useApiAction.js
  useModelParams.js
lib/
  models.js
  format.js
pages/
  CalibratePage.jsx
  SimulatePricePage.jsx
  RiskPage.jsx
  RunsPage.jsx
components/
  Layout.jsx
  TickerSelector.jsx
  ParamDisplay.jsx
  ParamInputs.jsx
  ModelSelector.jsx
  ConfidenceCone.jsx
  PayoffHistogram.jsx
  GreeksTable.jsx
  RiskGauges.jsx
  ScenarioComparison.jsx
  RunsTable.jsx
  LoadingSpinner.jsx
  ui.jsx
```

30 files total. Dependencies: react, react-dom, react-router-dom, recharts, axios, tailwindcss, @tailwindcss/vite, lucide-react.

9 Cumulative Milestone Status

Next: M8 — Docker + Infrastructure + Polish

Milestone	Status	Cumulative Tests
M1 — Data Ingestion		—
M2 — Calibration Engine		24
M3 — Simulation Engine		96
M4 — Options Pricing		183
M5 — Risk Analytics		345
M6 — FastAPI Backend		487
M7 — React Frontend		487 (no FE tests)